

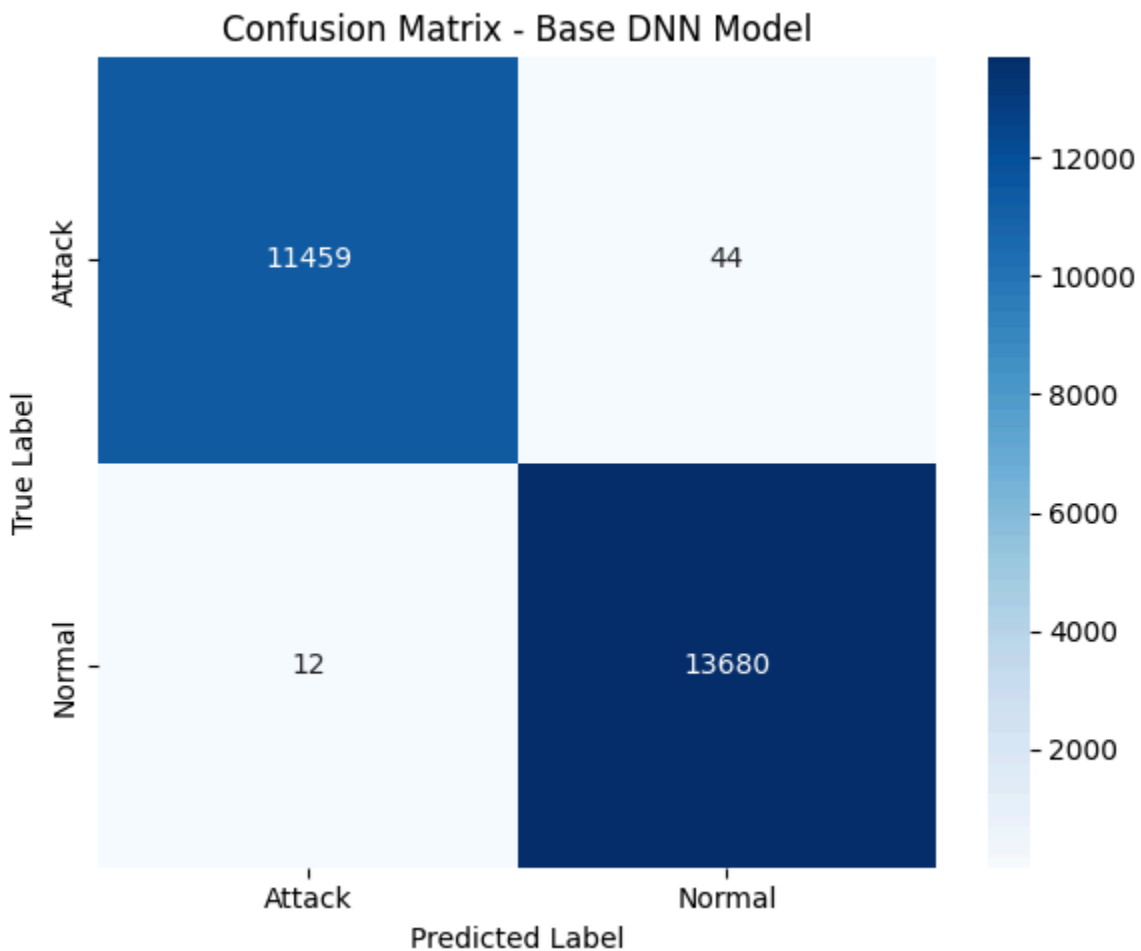
EE446
HW2
Discussion Questions

Q3)

c) Baseline model metrics

Classification Report:

	precision	recall	f1-score	support
Attack	1.00	1.00	1.00	11503
Normal	1.00	1.00	1.00	13692
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

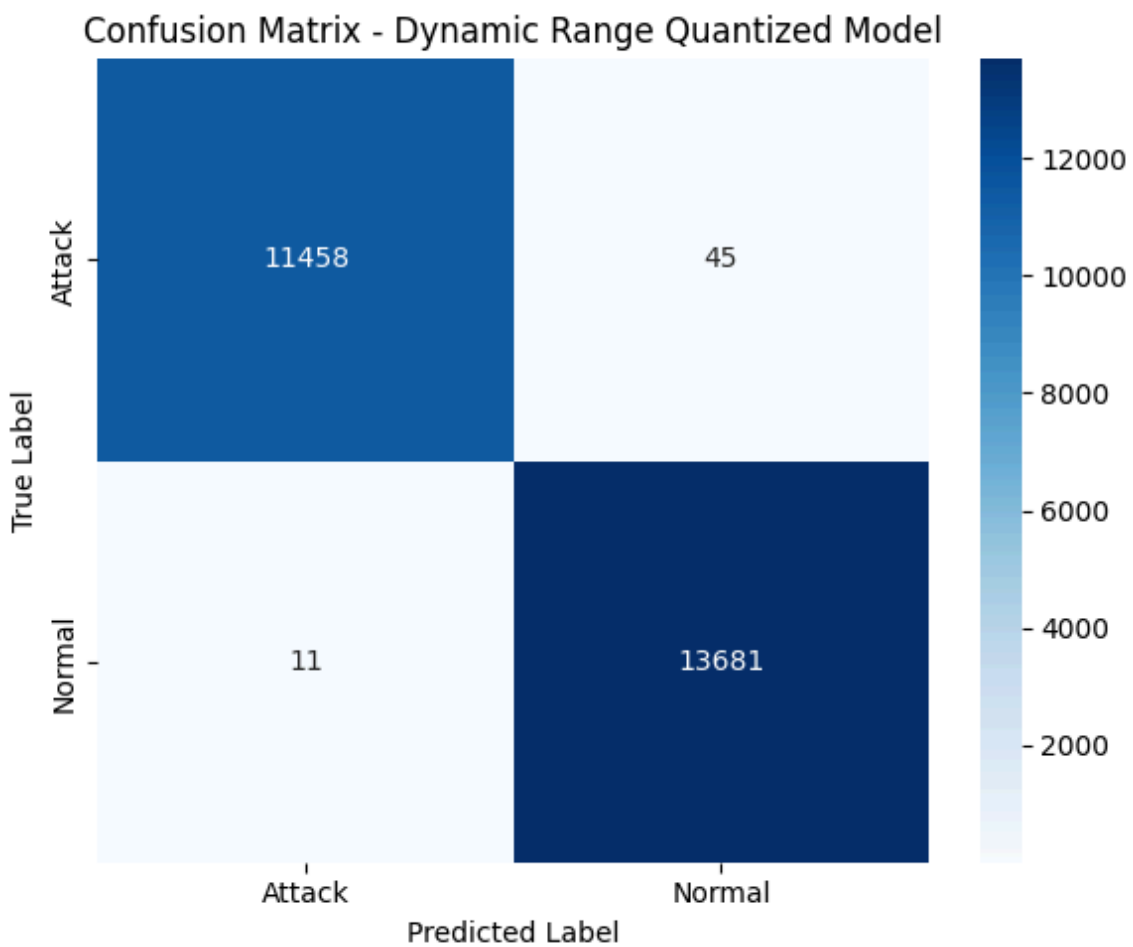


Q4)

b) Quantised model metrics

Classification Report (Dynamic Range Quantized Model):

	precision	recall	f1-score	support
Attack	1.00	1.00	1.00	11503
Normal	1.00	1.00	1.00	13692
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195



Q5)

Dynamic range quantization caused a TensorFlow Lite Micro / Arduino compatibility error on my setup. Therefore, I implemented full integer int8 quantization as the deployment-compatible workaround.

c) Printed Serial Monitor outputs for five samples

```
17:08:13.406 -> Predicted Class: Attack
17:08:13.406 -> Actual Class: Normal
17:08:13.406 -> -----
17:08:24.414 -> Sample #: 1
17:08:24.414 -> Predicted Class: Attack
17:08:24.414 -> Actual Class: Normal
17:08:24.414 -> -----
17:08:25.416 -> Sample #: 2
17:08:25.416 -> Predicted Class: Attack
17:08:25.416 -> Actual Class: Normal
17:08:25.416 -> -----
17:08:26.387 -> Sample #: 3
17:08:26.387 -> Predicted Class: Attack
17:08:26.422 -> Actual Class: Attack
17:08:26.422 -> -----
17:08:27.425 -> Sample #: 4
17:08:27.425 -> Predicted Class: Attack
17:08:27.425 -> Actual Class: Attack
17:08:27.425 -> -----
17:08:28.391 -> Sample #: 5
17:08:28.391 -> Predicted Class: Attack
17:08:28.391 -> Actual Class: Normal
17:08:28.429 -> -----
```

d) Printed Serial Monitor outputs for ten samples excluding the first five samples

```
-----
17:28:06.326 -> Sample #: 1
17:28:06.326 -> Predicted Class: Attack
17:28:06.326 -> Actual Class: Normal
17:28:06.326 -> -----
17:28:07.296 -> Sample #: 2
17:28:07.296 -> Predicted Class: Attack
17:28:07.296 -> Actual Class: Normal
17:28:07.296 -> -----
17:28:08.323 -> Sample #: 3
17:28:08.323 -> Predicted Class: Attack
17:28:08.323 -> Actual Class: Attack
17:28:08.323 -> -----
17:28:09.339 -> Sample #: 4
17:28:09.339 -> Predicted Class: Attack
17:28:09.339 -> Actual Class: Normal
17:28:09.339 -> -----
17:28:10.302 -> Sample #: 5
17:28:10.302 -> Predicted Class: Attack
17:28:10.338 -> Actual Class: Attack
17:28:10.338 -> -----
17:28:11.307 -> Sample #: 6
17:28:11.307 -> Predicted Class: Attack
17:28:11.340 -> Actual Class: Normal
17:28:11.340 -> -----
```

```
-----  
17:28:32.334 -> Sample #: 7  
17:28:32.334 -> Predicted Class: Attack  
17:28:32.334 -> Actual Class: Normal  
17:28:32.377 -> -----  
17:28:33.361 -> Sample #: 8  
17:28:33.361 -> Predicted Class: Attack  
17:28:33.361 -> Actual Class: Attack  
17:28:33.361 -> -----  
17:28:34.339 -> Sample #: 9  
17:28:34.339 -> Predicted Class: Attack  
17:28:34.339 -> Actual Class: Normal  
17:28:34.373 -> -----  
17:28:35.388 -> Sample #: 10  
17:28:35.388 -> Predicted Class: Attack  
17:28:35.388 -> Actual Class: Normal  
17:28:35.388 -> -----
```